

What is an AI system?

Most of what users call “the AI” is the system you build around the model.

“The model generates. The system does.”

Ask on every project

1. Which ring is this really: hand-written rules, classic ML or a generative model?
2. What exactly is sent to the model on each call, and who decides that?
3. Where do facts the model can't know come from: private data, recent events?
4. Which actions can it take, and who or what approves them before they run?
5. How will we know it got worse after a model, prompt or data change?

Rules of thumb

- Name the box before you fix anything: model, context, knowledge, tools or another.
- Rule out context, knowledge and tools before you blame the model.
- Use a plain rule instead of a model when a fixed rule does the job reliably.
- Judge quality on a set of test cases you can rerun, not on one good demo.

Red flags

- ⚠ “The AI decided” and nobody can say which part made the call
- ⚠ Nobody can list what goes into the model's context on a request
- ⚠ The model can take actions with no permission check or approval step
- ⚠ A launch plan with no way to measure quality after changes

Key terms

Model	Trained weights behind an endpoint: text in, predicted text out.
Inference	Running the trained model on a request. Its weights don't change.
Orchestration	The logic that decides which steps, model calls and tools run, and in what order.
Guardrails	Checks and limits around the model: filters, permissions, approvals.
Evaluation	Repeatable tests that show whether the system is good enough, and stays so.

CHECKLIST

System map in five minutes

- ☐ Pick one real user request and trace it from the interface to the answer, step by step.
- ☐ Mark the model call. Every other step belongs to the system.
- ☐ Sort each step into the eight boxes; one step can touch two boxes.
- ☐ For each empty box, ask: missing by design, or missing by accident?
- ☐ Note what each box adds per request in cost and delay.
- ☐ Circle the box most likely to fail first, and name who owns it.